

ENGR 1451/2451: Exploratory Data Science

More about Correlation

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Today's Goals

Use correlation carefully: know what changes it, what does not, and when it can mislead.

- Explain why correlation is unitless and unchanged by simple unit conversions.
- Predict how swapping variables, shifting, or rescaling affects r .
- Recognize outliers and nonlinear patterns where r is a poor summary.
- Distinguish individual-level correlations from ecological correlations.
- Explain why association alone does not establish causation.

Warm-Up: What Should Happen to r ?

A small data set has $r \approx 0.76$.

x	y
1	2
2	3
3	1
4	5
5	6

Question

What happens to r if we...

1. switch the two columns?
2. add 3 to every y value?
3. double every x value?
4. multiply every y value by -1 ?

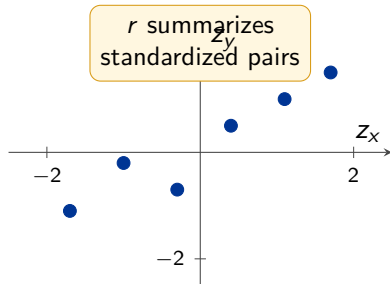
Answers: same, same, same, and sign flips to about -0.76 .

Correlation Is a Pure Number

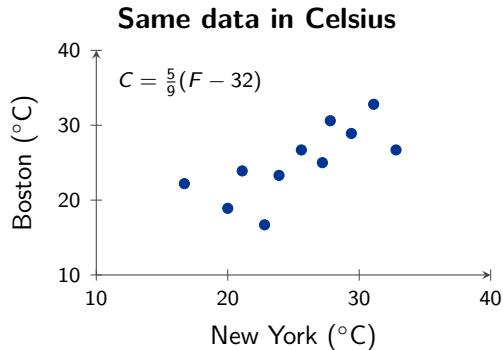
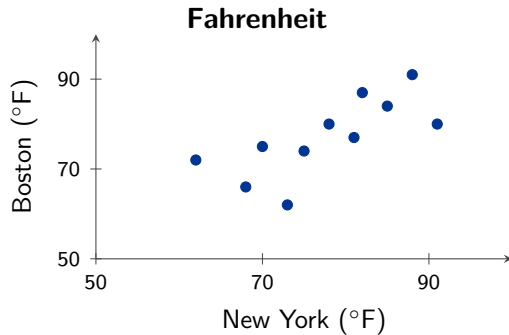
Correlation is computed after converting both variables to standard units:

$$z_x = \frac{x - \bar{x}}{s_x}, \quad z_y = \frac{y - \bar{y}}{s_y}.$$

Why units cancel: Adding a constant shifts the average too. Multiplying by a positive constant scales both deviations and the SD.

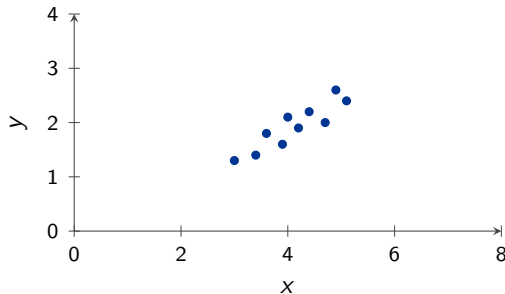


Unit Conversions Do Not Change r

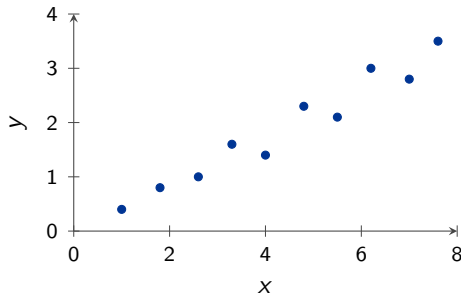


Same r Can Look Different

Narrow spread in both variables



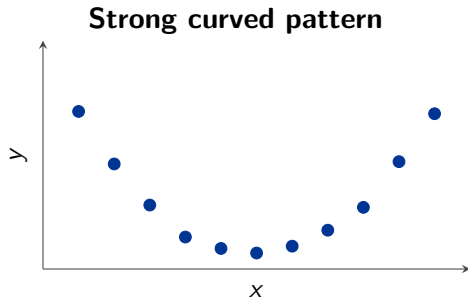
Wider spread in both variables



Idea

Correlation describes clustering **relative to the variables' spreads**, not in the original measurement units.

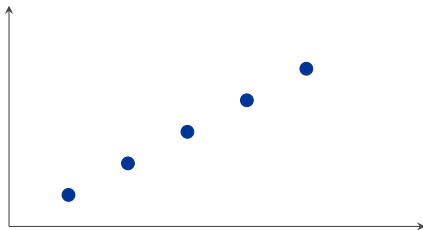
Correlation Measures Linear Association



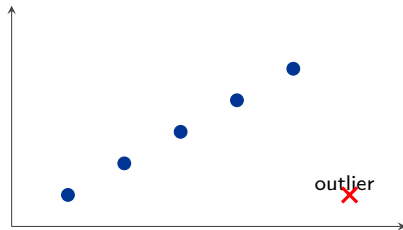
- r summarizes straight-line association.
- A curved pattern can have $r \approx 0$.
- Do not translate “small r ” into “no relationship” without seeing the plot.

Outliers Can Dominate r

Nearly perfect linear pattern



Same pattern plus one outlier



Better practice

Investigate unusual points. Remove a point only with a good reason, not just because it changes the answer.

Ecological Correlations

An **ecological correlation** is computed from group rates or group averages.

Individual level

Each point is a person, item, product, sample, etc.

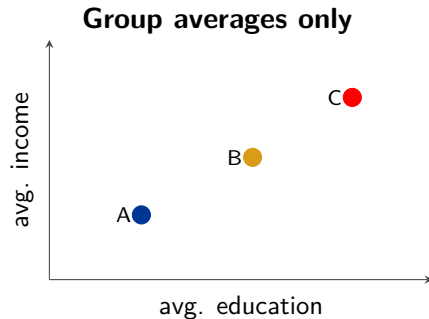
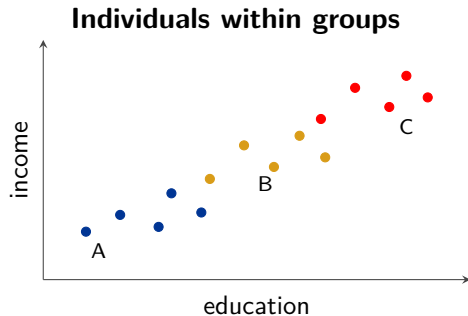
Group level

Each point is a state, country, classroom, year, school, company, etc.

- Group averages often look less scattered.
- The group-level correlation can be much larger.
- It may not estimate the individual-level relationship.

Same variables. Different unit of analysis.

Individuals Versus Group Averages



Watch out

Replacing individuals by averages removes much of the within-group scatter. Ecological correlations often overstate the strength of an association.

Quick Check: Ecological or Individual?

Question

For each data set, is the correlation ecological or individual-level?

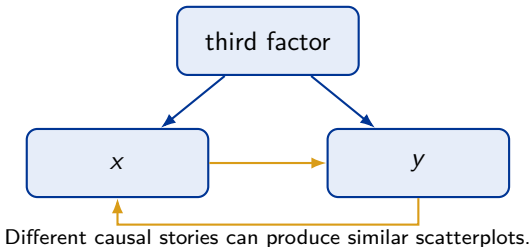
1. Each row is one student: study hours and exam score.
2. Each row is one class section: average study hours and average exam score.
3. Each row is one country: per-capita cigarette consumption and lung-cancer death rate.
4. Each row is one product: thickness and measured strength.

Answers: (1) individual, (2) ecological, (3) ecological, (4) individual/item-level.

Association Is Not Causation

Correlation measures association. It does not identify the mechanism.

- x may cause y .
- y may cause x .
- A third factor may affect both.
- The pattern may be an artifact of measurement, sampling, or grouping.

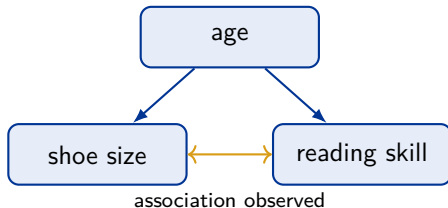


Confounding Example: Shoe Size and Reading

Among school children, shoe size and reading skill may be strongly associated.

Does reading make feet grow?

No. Age is a third factor: older children tend to read better and have larger feet.



Example: Education and Unemployment

During the Great Depression, better-educated job seekers tended to have shorter spells of unemployment.

Tempting conclusion

Education protected workers from unemployment.

Caution

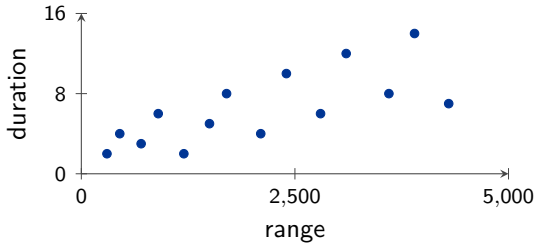
The data were observational. Age was a confounder: younger people tended to be better educated, and employers tended to prefer younger job seekers.

Controlling for age made the apparent education effect much weaker.

Example: Species Range and Duration

Observed association

For 99 gastropod species, geographical range and species duration had a positive association.



1. Wide range may help a species survive longer.
2. A long-lived species may have more time to spread.
3. Species with wider range may be easier to find in the fossil record.

The correlation alone does not choose among these explanations.

Example: Diet and Cancer Across Countries

Across countries, higher per-capita fat consumption was associated with higher breast-cancer death rates.

Why this is not enough

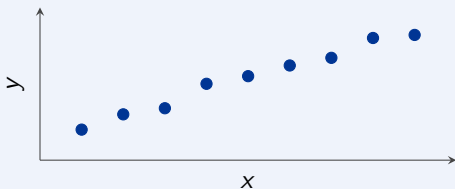
The points are countries, not people. Other country-level differences may move with diet.

- Richer countries may have more fat *and* more sugar.
- Lifestyle, screening, reporting, and other diet factors may differ.
- An ecological pattern can suggest a question, but does not settle causation.

Ask: what individual-level evidence or study design would be needed?

Causal Explanations to Consider

Observed association



Before claiming causation, ask:

1. What was the study design?
2. Could a third variable explain it?
3. Could the direction be reversed?
4. Could measurement, selection, or aggregation create the pattern?

Key Takeaways

- Correlation is unitless because it is based on standardized values.
- Switching variables, adding constants, and multiplying by positive constants do not change r .
- Correlation measures linear association; nonlinear patterns and outliers can make it misleading.
- Ecological correlations use group rates or averages and can overstate individual-level association.
- Association is not causation; look for confounders, reverse direction, and study-design limits.